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Polychronicity and the Inventory of Polychronic Values (IPV): The development of an instrument to measure a fundamental dimension of organizational culture

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Polychronicity and the Inventory of Polychronic Values (IPV)

Inventory of
Polychronic
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205

The development of an instrument to measure a fundamental dimension of organizational culture

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Abstract *The ten-item Inventory of Polychronic Values (IPV), a psychometric measure of polychronicity (the extent to which people in a culture prefer to be engaged in two or more tasks or events simultaneously and believe their preference is the best way to do things), was developed using data from 11 samples (N = 2,190) collected from bank employees, undergraduate students, hospital personnel, dentists and their staffs, and state agency managers. Principal components, alpha, correlation, and confirmatory factor analyses supported the IPV in its internal consistency, test-retest reliability, content adequacy, construct validity (both discriminant and convergent), and nomological validity.*

Over the past decade and a half, questions about organizational culture's impacts on many aspects of organization-level (e.g. Kotter and Heskett, 1992;

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Peters and Waterman, 1982) and individual-level (e.g. O'Reilly *et al.*, 1991; Sheridan, 1992) behaviors have been prominent organizational research topics. However, with the exception of an occasional reference to long-term versus short-term time horizons (e.g. Das, 1986; Ouchi, 1981), little of the organizational culture research has addressed what may be some of the most fundamental dimensions of any culture, the temporal dimensions. A few theorists (Bluedorn and Denhardt, 1988; McGrath and Rotchford, 1983; Schein, 1992) have discussed the temporal attributes of organizations and their cultures, but little empirical research – Schriber and Gutek's (1987) scale-development work being a notable exception – conducted within actual organizations has addressed the temporal dimensions. Even Schriber and Gutek's (1987) comprehensive temporal scale-development work did not include a scale for what may be the most basic temporal dimension of all, polychronicity, which concerns how many things an individual attends to and is involved with simultaneously (Hall and Hall, 1990).

Several recent studies suggest just how important polychronicity may be for understanding organizational phenomena. Eisenhardt (1989) found that the more alternatives considered simultaneously by strategic management teams in high velocity environments, the faster those teams made strategic decisions. Also, speed of strategic decision making was positively related to organizational performance in her data. Judge and Miller (1991) then replicated these findings for organizations in high velocity environments. Brown and Eisenhardt's (1997) subsequent research has examined the process of how organizations successfully manage multiple product development projects simultaneously. Finally, in organizations with matrix structures, McCollum and Sherman (1991) found significant positive correlations between the percentage of personnel assigned to two projects and measures of organizational performance, but no significant correlations between organizational performance and the percentage of personnel assigned to either just one project or to three or more projects. All of these studies dealt with behaviors that varied across the polychronicity spectrum, but none of them used this variable to conceptualize their work (or even mentioned it).

An exception to this practice is Schein's theoretical discussion of polychronicity. In both editions of his seminal work on organizational culture, Schein (1985, 1992) has argued that polychronic values are more suitable for small organizations, for "the early stages of an organization" (Schein, 1992, p. 108), whereas monochronic values are more suitable for large organizations. He believed that monochronic time is more suitable for situations requiring highly co-ordinated activities typically found in large organizations whereas polychronic time is better suited for complex problem solving requiring high levels of interaction and open communication channels, a context more typically found in small organizations. This interpretation is consistent with Eisenhardt's (1989) findings described earlier and her interpretation of them, a rationale implying that large monochronic and small polychronic organizations should be more effective than their opposites.

Because the investigation of polychronicity's role in organizational processes and behaviors has been severely hindered by the lack of a satisfactory measurement instrument, we decided to develop one. Thus, we conducted the present series of studies to develop a valid and reliable measure of polychronicity, the Inventory of Polychronic Values (IPV), as a dimension of organizational culture. In what follows we:

- (1) present a definition and explanation of polychronicity;
- (2) describe our original efforts to modify and improve the reliability of an earlier polychronicity scale, an effort that eventually produced the IPV;
- (3) quantitatively assess the IPV's content adequacy;
- (4) assess the IPV's construct validity; and
- (5) conduct a nomological validity test of the IPV.

We conclude our analysis with a discussion of our findings and their implications for organizational research.

Polychronicity

After first introducing the concept (Hall, 1959), Hall (1983, pp. 45-46) subsequently described polychronicity as a cultural variable involving two different ways of organizing activities:

- (1) monochronically – involvement in events one event at a time; and
- (2) polychronically – involvement in two or more events at the same time;

the monochronic and polychronic approaches being opposites (Hall and Hall, 1990, p. 13).

As a cultural variable then, we define polychronicity as the extent to which people in a culture:

- (1) prefer to be engaged in two or more tasks or events simultaneously; and
- (2) believe their preference is the best way to do things.

This definition taps both the values ("preferences") and beliefs ("believe ...") attributes included in many definitions of culture (see Kroeber and Kluckhohn, 1952). Bluedorn *et al.* (1992) have noted that the element of "at the same time" or "simultaneously" in the polychronicity concept can refer literally to "at the same time", such as an individual eating lunch and reading a newspaper simultaneously, as well as to the active interspersing and dovetailing of several activities within the same time period, such as a morning's work. In extremely monochronic cultures, people would tend to focus on a single task or project for the entire morning and regard unscheduled events (e.g. phone calls) as interruptions. Conversely, people in more polychronic cultures would anticipate involvement with multiple activities during the same morning period, intending to move back and forth among several tasks and projects during the morning. Unscheduled events such as phone calls, or colleagues and

customers dropping in without appointments, would be interpreted as part of the morning's normal activities rather than as interruptions or deviations from the plan or schedule[1].

Developing the Inventory of Polychronic Values

Kaufman *et al.* (1991) developed the Polychronic Attitudes Index (PAI), a four-item scale to investigate individual differences in polychronicity values and behaviors. Although the alpha coefficient they reported, 0.68, was lower than Nunnally's (1978, p. 245) recommended 0.80 for basic research, their work (Kaufman *et al.*, 1989, 1991) provided a pool of initial items that we modified and expanded to develop a scale measuring the extent to which an organization's culture is polychronic. Following Nunnally's (1978) recommendation, which was recently reaffirmed by Clark and Watson (1995), one of our goals was to develop a scale that consistently produced alpha coefficients in the 0.80 range or higher[1].

Sample 1

With an emphasis on representing all aspects (normative, preferences, both ends of the scale) of the polychronicity construct in reasonable proportions while still focusing exclusively on polychronicity (as recommended by Clark and Watson, 1995), we modified several of the items from Kaufman *et al.*'s (1991) scale and developed two completely new items. We included these six items in an organizational culture study of a medium-sized bank in a small mid-western city. We attempted to administer the questionnaire to the bank's entire full-time work force during work hours, and the 205 respondents represented 88.7 percent of the bank's full-time workforce, including the president and all members of the executive committee. This 205-respondent sample had a mean age of 33.1 years, a mean tenure at the bank of 5.2 years, and represented the different hierarchical levels as follows: middle and top managers, 16.1 percent; supervisory-level managers, 15.1 percent; and non-managerial personnel, 67.8 percent.

A principal components analysis of the data ($N = 205$) revealed that five of the proposed six items loaded on a single factor (all five loadings were ≥ 0.60) that explained 49.3 percent of the variance. When combined into a single scale, the five items produced an alpha coefficient (Cronbach, 1951) of 0.74, which was higher than the 0.68 alpha Kaufman *et al.* (1991) reported for their four-item PAI. However, Lautenschlager's (1989) ALPHATST program revealed this increase was not statistically significant ($\chi^2(1) = 1.68$, n.s.). The statistical tests performed by the ALPHATST program take into account:

- (1) the number of items in each scale;
- (2) the sample sizes;
- (3) the alpha magnitudes; and
- (4) whether the scales are from the same or from different samples (see Feldt *et al.* (1987) for a review of this family of statistical tests).

Sample 2

Based on results from the preliminary scale development effort in sample 1, we developed seven additional items with the intent of improving the scale's internal consistency while still maintaining our goal of representing the construct's content domain in reasonable proportions. Combining these items with items used in the sample 1 scale-development effort, we included 12 items in a questionnaire administered to 115 senior business majors in three business policy classes at a large public university located in the mid-western USA. The sample had a mean age of 21.9 years, and 52.2 percent of the respondents were men.

A principal components analysis revealed that ten of the 12 items (including all five of the items from the sample 1 scale) loaded on a single factor (all ten loadings were ≥ 0.55 , and five items loaded positively, five negatively, as anticipated) that explained 45.6 percent of the variance. A scree plot (Gorsuch, 1983) also supported a single-factor solution. The ten-item scale produced an alpha coefficient of 0.86, which was significantly higher, $\chi^2(1) = 18.06$, $p < 0.001$, than the 0.68 alpha reported by Kaufman *et al.* (1991) for their PAI scale and also significantly higher, $\chi^2(1) = 9.63$, $p < 0.003$, than the alpha of 0.74 produced by the five-item scale in sample 1. The respondents in this sample also completed the Marlowe-Crowne Social Desirability Scale (Crowne and Marlowe, 1964), and its correlation with the ten-item scale was only $r = 0.03$, n.s., indicating the ten-item scale did not seem to be subject to major social desirability response bias effects. At this point we designated the scale the IPV. The IPV's ten items are presented in Table I, along with the response format.

Samples 3, 4, 5, and 6: test-retest reliability

We assessed the IPV's test-retest reliability over four intervals: a nearly instantaneous retest interval of one-to-two minutes, and intervals of one, two, and four weeks. The samples for the nearly instantaneous (sample 3, $N = 24$, mean age = 22.46 years, 66.7 percent were men) and one-week intervals (sample 4, $N = 33$, mean age = 21.91 years, 54.3 percent were women) consisted of senior undergraduate business majors in two business policy courses, respectively, who responded to the scale in reference to either a current or recent former employer. The samples for the two- and four-week intervals (sample 5, $N = 23$, mean age = 36.79 years, 52.2 percent were women; and sample 6, $N = 21$, mean age = 35.02 years, 61.9 percent were women, respectively) were full-time managers in the Missouri state government who were enrolled in two Master of Public Administration (MPA) courses, respectively. They responded to the IPV about their agencies in the state government. We collected these four samples solely to assess the IPV's test-retest reliability, so aside from a few demographic items, we collected no other data besides IPV responses from these respondents.

The test-retest correlations from the four samples are as follow: for the immediate retest, $r = 0.95$; for the one-week retest interval, $r = 0.82$; for the two-week interval, $r = 0.89$; and for the four-week interval, $r = 0.78$. The magnitudes

Item	Sample 2 M SD	Sample 3 M SD	1	2	3	4	5	6	7	8	9	10		
1. We like to juggle several activities at the same time	5.62	1.30	3.78	1.80	–	0.40	0.44	0.16	0.36	0.26	0.37	0.32	0.28	0.27
2. We would rather complete an entire project every day than complete parts of several projects (R)	4.09	1.84	2.75	1.62	0.12	–	0.20	0.24	0.32	0.25	0.47	0.19	0.41	0.42
3. We believe people should try to do many things at once	4.80	1.63	3.38	1.81	0.57	0.09	–	0.46	0.42	0.62	0.42	0.51	0.36	0.45
4. When we work by ourselves, we usually work on one project at a time (R)	4.39	1.83	2.90	1.68	0.22	0.19	0.31	–	0.45	0.37	0.42	0.34	0.35	0.51
5. We prefer to do one thing at a time (R)	3.45	1.63	2.91	1.73	0.41	0.19	0.43	0.51	–	0.21	0.56	0.35	0.55	0.48
6. We believe people do their best work when they have many tasks to complete	4.30	1.42	3.03	1.66	0.40	0.14	0.38	0.11	0.28	–	0.42	0.47	0.32	0.45
7. We believe it is best to complete one task before beginning another (R)	3.84	1.51	2.82	1.53	0.34	0.23	0.43	0.40	0.46	0.34	–	0.29	0.65	0.59
8. We believe it is best for people to be given several tasks and assignments to perform	4.90	1.33	3.63	1.72	0.37	0.14	0.33	0.14	0.22	0.41	0.30	–	0.35	0.41
9. We seldom like to work on more than a single task or assignment at the same time (R)	3.51	1.54	3.52	1.59	0.35	0.20	0.24	0.31	0.42	0.24	0.40	0.24	–	0.45
10. We would rather complete parts of several projects every day than complete an entire project	4.29	1.52	3.05	1.69	0.27	0.30	0.30	0.01	0.24	0.32	0.17	0.32	0.06	–
Scale/10	4.47	1.05	3.18	1.01										
Sample size (N)	109		201											
Coefficient alpha (Cronbach, 1951)	0.86		0.80											

Notes: Items marked with (R) were reverse scored. Each item was scored on a seven-point Likert-type scale with 7 = strongly agree, 6 = moderately agree, 5 = slightly agree, 4 = neither agree or disagree, 3 = slightly disagree, 2 = moderately disagree, and 1 = strongly disagree. The sample sizes are based on listwise deletion of missing data. Items 2, 6, 8, and 10 were modified slightly in sample 3 to make them more specific to the dental sample (e.g. project was changed to procedure). Items 4 and 10 in sample 3 also used the word “task” rather than “project”, and item 6 used the phrase “to do” rather than “to complete”.

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of these test-retest correlation coefficients compared favorably with the magnitudes of the test-retest coefficients produced in research on other scales. For example, a search of Robinson *et al.*'s (1991) collection of scales revealed test-retest data about 47 studies that used a retesting interval of two weeks or less. These test-retest coefficients ranged from $r = 0.19$ to $r = 0.92$, with a median value of $r = 0.75$ and a mean, calculated via Fisher's (1921) r -to- z transformation, of 0.75. All four of the test-retest coefficients we obtained exceeded the average value of 0.75 in the scales reviewed and presented in Robinson *et al.* (1991).

Collecting these data also provided eight more alpha coefficients (two per sample with the initial scale administration reported first), which were 0.76 and 0.87 for the immediately retested sample, 0.82 and 0.91 for the sample with a one-week retest interval, 0.84 and 0.84 for the sample with a two-week retest interval, and 0.84 and 0.93 for the sample with a four-week interval. At this point in our scale development work, we had obtained alpha coefficients from nine administrations of the ten-item scale (these eight and the administration in sample 2). The median of these nine alphas was 0.84.

The highly satisfactory exploratory factor analysis results and internal consistency and test-retest reliability assessments concluded this phase of our scale development work. The next phase concerned content adequacy, which is presented next.

Assessing the IPV's content adequacy (validity)[2]

The Q-method approach

Schriesheim *et al.* (1993) defined content adequacy as the degree to which a measure's items are a proper sample of the theoretical content domain of a construct, and we have followed their recommendation to use the term "adequacy" rather than "validity" due to usage criticisms of the latter in this context. Further, they emphasized the strategic importance of a measure containing a representative sample of the underlying construct's theoretical universe, perhaps even in correct proportion to a construct's subdomains. They note that content adequacy has historically been assessed subjectively rather than through quantitative analytical procedures. To provide a quantitative approach, Schriesheim *et al.* (1993) developed the Q-method approach based on Stephenson's (1953) Q-methodology. This is the approach to content-adequacy assessment that we employed.

The technique requires judges to complete a questionnaire containing two or more scales or subscales. The first scale's definition is presented to the judges who are then asked to rate each item according to how much it refers to the construct whose definition was presented. Each item from the intended scale as well as from the other scales is assessed this way. The process is then repeated with the definitions of each of the other scales, so each judge evaluates each individual scale item as many times as there are scale definitions – and then once more because the final portion of the questionnaire asks each judge to rate each item according to how much it refers to something other than any of the

definitions provided for the preceding ratings. Results of these ratings are then compared. A detailed description of this method is provided in Schriesheim *et al.* (1993).

To assess the IPV's content adequacy with this method, we included two of Schriber and Gutek's (1987) scales – schedules-and-deadlines emphasis (nine items) and emphasis on punctuality (four items) – along with the IPV's ten items. Thus, each subject/judge would evaluate each of the 23 items four times: once for each construct's definition and once for the extent to which each item referred to something other than any of the three constructs' definitions. We chose the schedules-and-deadlines and punctuality scales because all three scales (counting the IPV) are scales of temporal values. Further, Hall (1983) observed that monochronic cultures take appointments and the adherence to schedules more seriously than polychronic cultures, suggesting that values emphasizing polychronicity should be related to values emphasizing schedules, deadlines, and punctuality. Thus, the items to be rated come from scales that should be intercorrelated, making the rating results a meaningful assessment.

We obtained usable ratings from sample 7, 68 students from three undergraduate courses (organizational behavior, human resources management, and business policy) at a large public mid-western university, which is 18 more judges than the 50 judges Schriesheim *et al.* (1993) suggested as a sufficient number. Their mean age was 22.04 years and 57 percent were men. The results of their ratings are presented in Table II.

Before interpreting the results in Table II, we checked the reliability of the judges' responses by randomly dividing the group of judges in half and calculating the Pearson product-moment correlations between the two groups for each of the four content categories. The correlations were 0.99, 0.98, 0.98, and 0.73 for the polychronicity, schedules-and-deadlines, punctuality, and something-else/none-of-these categories, respectively. All of these are satisfactorily high, except the 0.73 for the something else/none-of-these correlation. However, this correlation was produced by two variables whose standard deviations were 0.19 and 0.23 compared to standard deviations ranging from 1.1 to 1.5 for the variables that produced the other three correlations. With such small standard deviations, a correlation of 0.73 is actually remarkably high. Thus, the judges' data appeared satisfactorily reliable, and we proceeded to interpret the data shown in Table II.

To support a scale's content adequacy, the items from the scale of interest should:

- (1) receive their highest ratings when the ratings are based on the scale-of-interest's definition (e.g. examine rows 1-10 in Table II);
- (2) receive higher ratings than the items from the other scales when the ratings are based on the scale-of-interest's definition (e.g. column 1 in Table II);
- (3) receive lower ratings than the items from the other scales when the ratings are based on the other scales' definitions (e.g. examine columns 2 and 3 in Table II); and

Row number and item	(1) Polychronicity definition	(2) Schedules- and-deadlines definition	(3) Punctuality definition	(4) Something- else/none-of- these values
1. Polychronicity 1	4.96	1.53	1.28	1.18
2. Polychronicity 2	4.13	2.13	1.65	1.34
3. Polychronicity 3	4.47	1.37	1.21	1.35
4. Polychronicity 4	3.97	1.76	1.43	1.50
5. Polychronicity 5	4.49	1.62	1.48	1.13
6. Polychronicity 6	4.13	1.29	1.28	1.31
7. Polychronicity 7	3.88	2.06	1.49	1.29
8. Polychronicity 8	4.13	1.46	1.31	1.34
9. Polychronicity 9	3.93	1.41	1.49	1.24
10. Polychronicity 10	3.82	1.88	1.41	1.47
11. Schedule-and-deadline 1	1.37	4.43	3.96	1.29
12. Schedule-and-deadline 2	1.40	4.71	4.01	1.28
13. Schedule-and-deadline 3	1.56	4.76	4.51	1.26
14. Schedule-and-deadline 4	1.50	4.40	3.60	1.28
15. Schedule-and-deadline 5	1.31	4.29	4.12	1.24
16. Schedule-and-deadline 6	1.40	4.75	3.43	1.47
17. Schedule-and-deadline 7	1.37	4.41	4.47	1.28
18. Schedule-and-deadline 8	1.46	4.59	3.93	1.51
19. Schedule-and-deadline 9	1.57	4.07	3.33	1.52
20. Punctuality 1	1.38	3.69	3.99	1.31
21. Punctuality 2	1.15	3.24	3.68	1.41
22. Punctuality 3	1.16	3.16	3.69	1.54
23. Punctuality 4	1.24	3.07	3.38	2.12

Notes: All items were rated with the following scale: 5 = completely or almost completely; 4 = much; 3 = moderately or about half; 2 = some; and 1 = none or hardly any. The polychronicity 1-10 items are the same items, in the same order, as the ten IPV items in Table I. The schedule and deadline 1-9 items and the punctuality 1-4 items are the Schriber and Gutek (1987) schedules-and-deadlines and punctuality scales, respectively, and the items are arranged in the same order in which Schriber and Gutek (1987) presented them on p. 646 of their article

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Table II.
Judge panel total
evaluations

- (4) receive higher scores when the ratings are based on the scale of interest's ratings than when the ratings concern the extent to which the items are about things other than any of the constructs defined in the questionnaire (e.g. examine rows 1-10 in columns 1 and 4 in Table II).

By these criteria, the data in Table II strongly support the IPV's content adequacy. Each of the ten IPV items had its highest rating when the definition of polychronicity was used as the referent, ratings that were higher than the schedules-and-deadlines and punctuality items and higher than the "none-of these/about something-else" ratings. The IPV items also received lower ratings than the schedules-and-deadlines and punctuality items when those scales' definitions were the basis of the ratings. Indeed, the magnitude of these

differences is greater than the magnitudes Schriesheim *et al.* (1993, p. 401) presented in their example of this technique.

Although the pattern of ratings associated with the ten IPV items reveals almost textbook support for its content adequacy, the schedules and deadlines and punctuality scales reveal potential problems. Although only one of the ratings “cross-loaded” for these two scales’ items, the ratings are often uncomfortably close, indicating the judges did not perceive major differences between these two constructs. These results illustrate the method’s ability to detect such differences (and lack thereof), which further strengthens these results’ support of the IPV’s content adequacy.

With the IPV’s content adequacy strongly supported by the Q-method, we shifted our concern to a second content adequacy test.

A known-groups test

A known-groups test for content adequacy compares scale scores for groups whose levels on the variable of interest are already known by other means. If the scale successfully differentiates the groups in the manner already known, it demonstrates content adequacy (Kerlinger, 1973, p. 495). To perform a known-groups test of the IPV, we needed to measure the polychronicity levels of different organizations with some instrument other than the IPV, then apply the IPV to those same organizations and compare the two sets of scores. A high level of similarity would indicate strong support for the IPV’s content adequacy whereas a small or non-existent degree of similarity would indicate a lack of content adequacy.

The difficulty of performing such a test on the IPV is the absence of any other psychometric measures of polychronicity as a cultural attribute. Indeed, their absence is the reason we decided to develop such a measure. Hall’s previous empirical research on polychronicity in national cultures, for example, was all based on qualitative methods and observations. To undertake comparable qualitative measurements to obtain judgments of the polychronicity levels in a large enough sample of organizations (i.e. $N = 30-50$) to provide a meaningful correlation with IPV scores from the same organizations would require years, perhaps a lifetime to produce. Consequently, we approached the problem by creating organizations with known polychronicity levels.

The “organizations” we created consisted of two descriptions of managerial behaviors and attitudes in organizations, one describing a very polychronic organization, the other a very monochronic organization. Care was taken to avoid any of the phrasing from the ten IPV items or even comparably synonymous phrasing in these descriptions so that the application of the IPV to these scenarios would provide a meaningful known-groups content adequacy test. In other words, we avoided statements in the scenarios such as “Jane is working on several projects at the same time” because a respondent applying the IPV to a scenario with such statements would simply be identifying scale

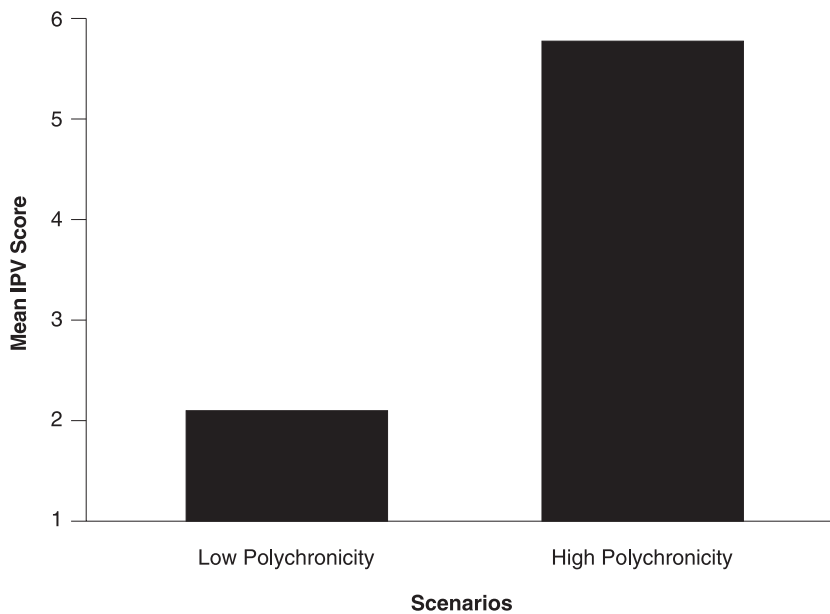
items in the scenario rather than using the IPV to judge polychronicity values. Thus, the scenarios and the IPV are lexically independent. The scenarios are presented in the Appendix.

As Rosenthal and Rosnow (1991, p. 61) noted, sometimes the validity of the criterion variables must be demonstrated, perhaps through other judgments. Because we created the scenarios for use in this known-groups test, we recognized the importance of demonstrating the polychronicity levels represented in the two scenarios by means other than our own assertions. So we identified ten scholars (faculty and doctoral students) in the USA who were already knowledgeable about polychronicity and asked each of them to evaluate on a ten-point scale (1 being extremely low polychronicity and 10 being extremely high polychronicity) the level of polychronicity represented in the two scenarios. Five of these expert judges read the intended high polychronicity scenario first whereas the other five judges read the intended low polychronicity scenario first. All ten expert judges rated the intended high polychronicity scenario as much more polychronic than the intended low polychronicity scenario. The means for the high and low polychronicity scenarios were 9.20 and 2.10, respectively, a statistically significant (paired difference *t*-test, $t(9) = 14.075, p < 0.001$) mean difference of 7.1 scale points on the ten-point scale. These results confirmed our original intent and judgments: one scenario describes an extremely polychronic organization whereas the other scenario describes an extremely monochronic organization.

Having strongly supported the known polychronicity levels of the two “organizations,” we administered the scenarios to sample 8, 285 students (mean age = 21.38 years and 52 percent were men) in a large introductory management course at a large public mid-western university. The instructions asked the students to read the scenario they were given (approximately half of the students were given the high polychronicity scenario while the other half were given the low polychronicity scenario) and to then complete the IPV in the same way they believed the managers described in the scenario would complete it. The results are presented in Figure 1.

As Figure 1 indicates, the mean IPV score for the high polychronicity scenario was 5.78; for the low polychronicity scenario, 2.10, a highly significant ($t(284) = 30.609, p < 0.001$) mean difference of 3.68 on the IPV’s seven-point scale (the ten items were summed and divided by ten, coefficient alpha = 0.97). In correlational terms, these results are an *r* and an eta of 0.88 (*r* and eta being identical in this case because the scenario variable is a dichotomy). Moreover, we conducted a more fine-grained analysis at the level of individual scale items. A MANOVA revealed a significant difference between the sets of scale items for the two scenarios (Wilks’ lambda = 0.202; $df = 10; p < 0.001$), and each item revealed a significant difference (univariate ANOVAs, $p < 0.001$) in the predicted directions between the two scenarios. Thus, the similarity of scores between the known groups (the two scenarios, one high polychronicity and the other low polychronicity) and the scale being assessed, the IPV, was very high thereby yielding a second successful test of the IPV’s content adequacy. Both

Figure 1.
Mean inventory of
polychronic values (IPV)
scores for high and low
polychronicity scenarios



assessments of the IPV's content adequacy produced results strongly supporting the IPV, so we continued our efforts and assessed its construct validity.

Assessing the IPV's construct validity

Construct validity addresses the question of what property or properties account for the variation in a test or scale (Kerlinger, 1973, p. 461), of what constructs account for variance in test performance (Cronbach and Meehl, 1955, p. 281). To address this question, we divided our efforts into an assessment of the IPV's homogeneity, and of its discriminant and convergent validity, gathering a large ($N = 1,416$) pool of data from three samples, samples 9, 10, and 11, with which to conduct these construct validity assessments. We analyzed all three of these samples to assess the IPV's homogeneity, and then used samples 10 and 11 to assess its discriminant and convergent validity. The three samples are briefly described next, followed by our analyses of data from them concerning the IPV's homogeneity, and discriminant and convergent validity.

The three samples

Sample 9. The sample consisted of 75 dentists whose participation was requested at the annual meeting of a state dental society (about 75 percent agreed to participate) and 141 of their dental practice employees (e.g. hygienists, receptionists, etc.). The mean age of sample members was 40.6 years, and mean tenure with the dental practices was 9.5 years. The sample

consisted of 62.5 percent women, and missing values subsequently reduced the effective sample size to 201.

Sample 10. We included the IPV in an extensive organizational culture assessment of a major hospital located in a large mid-western city. The questionnaires used in the culture assessment were completed during work hours by approximately 60 percent of the hospital's full-time workforce, including all of the top-management team. A total of 982 respondents completed questionnaires at the hospital. The sample's mean age was 39.6 years, and the mean tenure at the hospital was 17.7 years. Women comprised 78.6 percent of the sample, and missing values on scale items subsequently reduced the effective sample size to 883.

Sample 11. The hospital from which we collected sample 10 was the larger member of a recently formed two-hospital system. We gave the same questionnaire to full-time employees at the second hospital in the system, and 367 respondents completed questionnaires during work hours (sample 11). The mean age of sample members was 40.6 years, and the mean tenure at the hospital was 17.9 years. Women comprised 78.5 percent of the sample, and missing values on scale items subsequently reduced the effective sample size to 332.

Homogeneity/unidimensionality

Homogeneity (or unidimensionality) indicates whether a scale's items assess a single underlying factor or construct (Clark and Watson, 1995, p. 315). If each item in a scale is interpreted as a different source or measure of the construct it is designed to measure, evidence of scale homogeneity can be interpreted as a form of convergent validity. To assess the IPV's homogeneity, we subjected the data from samples 9, 10, and 11 to confirmatory factor analyses (CFAs) to identify the most parsimonious factor structure (number of factors) that provided a good fit to the data. We then conducted a test of partial model invariance across the three samples to examine the stability of both the factor structure and the pattern of factor loadings.

We should note here that all CFAs reported in this article were performed with EQS, version 5.6, and used the respective samples' covariance matrices, although as Hoyle and Panter (1995) recommend, correlation matrices are reported in Tables I and III for greater ease of interpretation. To evaluate the results of the CFAs undertaken in this research, we followed Hoyle and Panter's (1995) recommendations and used:

- (1) chi-square significance tests;
- (2) the goodness-of-fit index (GFI);
- (3) one type-2 index – the incremental fit index (IFI); and
- (4) one type-3 index – the comparative fit index (CFI).

We also followed Hu and Bentler's (1995) prescription to always report the standardized root mean square residual (SRMR) due to its direct

Table III.

Scale item means, standard deviations, and intercorrelations for samples 10 (below the diagonal) and 11 (above the diagonal)

Item	Sample 10		Sample 11		1	2	3	4	5	6	7	8	9	10
	M	SD	M	SD										
1. We like to juggle several activities at the same time	5.31	1.44	5.29	1.41	-	-0.00	0.53	0.17	0.28	0.22	0.17	0.32	0.12	0.16
2. We would rather complete an entire project every day than complete parts of several projects (R)	3.99	1.73	3.63	1.68	0.09	-	0.10	0.18	0.27	0.19	0.52	0.23	0.17	0.30
3. We believe people should try to do many things at once	4.99	1.68	5.06	1.76	0.48	0.15	-	0.21	0.25	0.42	0.23	0.47	0.13	0.31
4. When we work by ourselves, we usually work on one project at a time (R)	4.10	1.66	4.30	1.65	0.05	0.37	0.15	-	0.43	0.17	0.32	0.11	0.21	0.10
5. We prefer to do one thing at a time (R)	4.56	1.69	4.67	1.66	0.29	0.37	0.33	0.50	-	0.21	0.35	0.25	0.29	0.18
6. We believe people do their best work when they have many tasks to complete	4.36	1.65	4.29	1.63	0.27	0.05	0.42	0.09	0.20	-	0.29	0.46	0.22	0.33
7. We believe it is best to complete one task before beginning another (R)	4.18	1.69	4.04	1.62	0.21	0.41	0.33	0.36	0.49	0.30	-	0.29	0.36	0.32
8. We believe it is best for people to be given several tasks and assignments to perform	4.95	1.51	4.81	1.60	0.36	0.11	0.44	0.07	0.22	0.46	0.30	-	0.20	0.42
9. We seldom like to work on more than a single task or assignment at the same time (R)	4.40	1.58	4.32	1.65	0.13	0.17	0.15	0.21	0.33	0.11	0.35	0.11	-	0.12
10. We would rather complete parts of several projects every day than complete an entire project	4.41	1.67	4.20	1.70	0.30	0.30	0.34	0.18	0.29	0.28	0.35	0.32	0.10	-
Scale/10	4.53	0.95	4.46	0.94										
Sample size (N)	883		332											
Coefficient alpha (Cronbach, 1951)	0.78		0.78											

Notes: Items marked with (R) were reverse scored. Each item was scored on a seven-point Likert-type scale with 7 = strongly agree, 6 = moderately agree, 5 = slightly agree, 4 = neither agree nor disagree, 3 = slightly disagree, 2 = moderately disagree; and 1 = strongly disagree. The sample sizes are based on listwise deletion of missing data

interpretability and Bentler's (1995) belief that the SRMR discriminates between correctly specified and mis-specified models better than any of the other fit indices. To interpret the GFI, IFI, and CFI, we employed the criterion of a value ≥ 0.9 as indicative of acceptable fit based on the precedent established and advocated by Bentler and Bonett (1980) and Bentler (1992) for these indices. To interpret the SRMR, less consensus exists for a criterion value; therefore, based on Hu and Bentler's (1995) discussion of the SRMR, we interpreted values ≤ 0.05 as indicating a good-fitting model.

We initially conducted three CFAs on the IPV data from each sample. Because the IPV is symmetrically counterbalanced (five items worded to indicate a very polychronic emphasis and five items worded to indicate a very low polychronic emphasis), we estimated three two-factor models to examine the effects of reverse scoring half the items. To do so, we employed the procedure used by Spector *et al.* (1997) to examine such effects. Specifically, without reverse scoring any items, we estimated three models with:

- (1) the five non reverse scored items constrained to load on the first factor; and
- (2) the five items designed for reverse scoring constrained to load on a second factor.

The three models differed in the specifications for the covariance between the two factors. In one model the covariance was set to zero (an orthogonal model), in another it was estimated freely (an oblique model), and in the third it was set to -1 (the equivalent of a one-factor solution). In all three models the individual items' error term covariances were set to zero.

Spector *et al.*'s (1997) procedures indicate the oblique factor model should fit the data the best of the three models if a reverse scoring effect is present rather than the effects of a second theoretical construct. In all three samples this proved true as the orthogonal model's statistics indicated the best fit of the three models (sample 9: $\chi^2(34) = 70.70$, CFI = 0.91, IFI = 0.92, GFI = 0.94, SRMR = 0.06; sample 10: $\chi^2(34) = 237.48$, CFI = 0.90, IFI = 0.90, GFI = 0.95, SRMR = 0.06; sample 11: $\chi^2(34) = 133.56$, CFI = 0.86, IFI = 0.87, GFI = 0.92, SRMR = 0.07). The oblique models in all three samples generally met or exceeded the criteria of acceptable fit established beforehand, albeit with an occasional near miss. However, these statistics reveal an unambiguous pattern of better fit than either the orthogonal (sample 9: $\chi^2(35) = 137.48$, CFI = 0.78, IFI = 0.79, GFI = 0.89, SRMR = 0.18; sample 10: $\chi^2(35) = 395.40$, CFI = 0.83, IFI = 0.83, GFI = 0.92, SRMR = 0.14; sample 11: $\chi^2(35) = 193.74$, CFI = 0.78, IFI = 0.79, GFI = 0.89, SRMR = 0.15) or the one-factor (factor covariance set to -1) models (sample 9: $\chi^2(35) = 116.99$, CFI = 0.83, IFI = 0.83, GFI = 0.89, SRMR = 0.08; sample 10: $\chi^2(35) = 615.73$, CFI = 0.72, IFI = 0.72, GFI = 0.84, SRMR = 0.10; sample 11: $\chi^2(35) = 247.96$, CFI = 0.71, IFI = 0.71, GFI = 0.85, SRMR = 0.09).

When we conducted a test of partial model invariance (the factor loadings were constrained to equality across samples but the factor covariances were

allowed to vary freely) across these three samples for the oblique model, the profile of fit statistics generated were quite good too (χ^2 (122) = 490.18, CFI = 0.89; IFI = 0.89; GFI = 0.93, SRMR = 0.08), especially allowing for the attempt to fit the same factor loadings to three large data sets.

Because alternative procedures have been suggested for testing reverse scoring effects (e.g. Alwin and Krosnick, 1985; Harvey *et al.*, 1985), we conducted a second test for a reverse scoring effect using the procedure recommended by Alwin and Krosnick (1985). This method involves the estimation of a model specified as follows:

- (1) the scale items are specified to load on the factors as directed by theory (one factor, polychronicity, in this case) but without recoding any of the reverse scored items;
- (2) all scale items are specified to load on a method factor and are constrained to have equal loadings on the method factor;
- (3) the covariance between the method factor and the theoretical factors is set to zero; and
- (4) the variances of all factors, theoretical and method, are set to one.

This model is then compared to a one-factor model in which all of the items are specified to load on a single factor. The results of these tests in all three samples are presented in Tables IV and V.

The change in chi-square values from the one-factor to the two-factor models were all significant at the $p < 0.001$ level in all three samples (the chi square differences being, each with 1 degree of freedom: 47.32, 396.34, and 104.2, respectively), indicating the reverse scoring model fit the data better in all three samples. The improvements shown in the other fit indices also parallel the findings obtained with Spector *et al.*'s (1997) procedures. When we tested this model for partial invariance (method loadings were constrained to equality within and across samples), the fit indices were again good; indeed, they were almost identical to those obtained with the partial model invariance test conducted earlier.

Overall, the scale revealed acceptable levels of fit as a bipolar measure of polychronicity, albeit with some of its variance attributable to its counterbalanced configuration. This extra variance, however, allows for:

- (1) a greater coverage of the polychronicity construct's content domain without recourse to statements that use the word "not"; and
- (2) protection against respondents' possible response-set patterns.

On balance, these benefits of the reverse scoring design seem to outweigh the additional scale variance it produces. These results indicate the IPV possesses a single theoretical factor, polychronicity, but one other potential theoretical factor must be examined before we can state this conclusion confidently.

Although we developed the IPV to measure a single underlying construct, polychronicity, we conducted additional CFAs to examine the possibility that

Model	χ^2	df	GFI	CFI	IFI	SRMR
<i>Sample 9 (N = 201)</i>						
Null	514.95	45	NA	NA	NA	NA
One-factor	116.99	35	0.89	0.83	0.83	0.08
Two-factor (polychronicity and reverse scoring)	69.67	34	0.94	0.92	0.93	0.06
Three-factor (normative, preference, and reverse scoring)	68.24	33	0.94	0.93	0.93	0.06
<i>Sample 10 (N = 883)</i>						
Null	2,115.23	45	NA	NA	NA	NA
One-factor	615.73	35	0.84	0.72	0.72	0.10
Two-factor (polychronicity and reverse scoring)	219.39	34	0.95	0.91	0.91	0.05
Three factor (normative, preference, and reverse scoring)	189.03	33	0.96	0.93	0.93	0.05
<i>Sample 11 (N = 332)</i>						
Null	773.94	45	NA	NA	NA	NA
One-factor	247.96	35	0.85	0.71	0.71	0.09
Two-factor (polychronicity and reverse scoring)	143.76	34	0.91	0.85	0.85	0.07
Three-factor (normative, preference and reverse scoring)	143.76	33	0.91	0.85	0.85	0.07

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Table IV.
Overall fit indices for the polychronicity models

Notes: GFI = goodness-of-fit index; CFI = comparative fit index;
IFI = incremental fit index; SRMR = standardized root mean squared residual

Item	Sample 9		Sample 10		Sample 11		Partial invariance	
	P	RS	P	RS	P	RS	P	RS
1.	1.21	0.41	0.70	0.49	0.61	0.45	0.74	0.47
2.	-0.45	0.41	-0.77	0.49	-0.79	0.45	-0.72	0.47
3.	1.25	0.41	1.04	0.49	1.03	0.45	1.08	0.47
4.	-0.83	0.41	-0.75	0.49	-0.67	0.45	-0.74	0.47
5.	-1.18	0.41	-1.14	0.49	-0.88	0.45	-1.09	0.47
6.	0.89	0.41	0.80	0.49	0.92	0.45	0.85	0.47
7.	-0.99	0.41	-1.15	0.49	-1.04	0.45	-1.09	0.47
8.	0.83	0.41	0.80	0.49	1.00	0.45	0.85	0.47
9.	-0.82	0.41	-0.58	0.49	-0.66	0.45	-0.63	0.47
10.	0.64	0.41	0.87	0.49	0.85	0.45	0.83	0.47
N	201		883		332		1,416	
χ^2	69.67		219.40		143.76		486.60	
(df)	(34)		(34)		(34)		(124)	
GFI	0.94		0.95		0.91		0.93	
CFI	0.92		0.91		0.85		0.89	
IFI	0.93		0.91		0.85		0.89	
SRMR	0.06		0.05		0.07		0.08	

Notes: P is the polychronicity factor, and RS is the reverse scoring factor. Unstandardized loadings are reported because EQS automatically releases equality constraints when calculating a standardized solution, and equality constraints were required for the Alwin and Krosnick (1985) scoring factor and the partial invariance test. Items 2, 4, 5, 7, and 9 were not reverse scored prior to these CFAs

Table V.
Unstandardized factor loadings and model fit statistics for the CFAs in samples 9, 10 and 11 and for the test of partial invariance across the three samples

two theoretical constructs might have been included in the scale. The wording of items 3, 6, 7, and 8 (see Tables I and III) was more explicitly normative (i.e. beliefs about how people should work) than the wording of items 1, 2, 4, 5, 9, and 10 (see Tables I and III), which was more focused on the way people like or prefer to work. Thus a three-factor model was specified with items 3, 6, 7, and 8 loading on a normative factor, items 1, 2, 4, 5, 9, and 10 loading on a preference factor (the other items were set to zero on the respective factors), and all items constrained to load equally on a reverse scoring factor. The covariances with the reverse scoring factor were set to zero.

The results are instructive. In sample 9 the chi-squares were not significantly different between the two models ($\chi^2(1) = 1.43$, n.s.), and the normative and descriptive factor correlation was $r = -0.95$ ($p < 0.001$). In sample 10, the chi-squares were significantly different ($\chi^2(1) = 30.36$, $p < 0.001$), but the correlation between the two theoretical factors was again robust: $r = -0.87$. Finally, in sample 11 the chi-squares were identical, so it would be impossible for them to be significantly different, and the correlation between the normative and descriptive factors was $r = -1.00$, $p < 0.001$. As Spector *et al.* (1997, p. 672) noted, two factors that are perfectly correlated are equivalent to a single factor, and these results produced one perfect correlation and two others so large that it would be appropriate to interpret them similarly.

These results indicate the specification of two theoretical factors fails to generally improve model fit, the one exception coming from a modest chi-square difference from a very large ($N = 883$) sample apt to produce such differences. Even this sample produced a very large correlation, $r = -0.87$, between the two theoretical factors. Therefore, both philosophy of science reasons (theoretical parsimony) and empirical reasons (extremely high correlations between the two theoretical factors in all three samples), support the one-theoretical-construct interpretation of the IVP over the two-theoretical-constructs interpretation.

Discriminant and convergent validity

Because Hall (1983) and Hall and Hall (1990) indicated that values for punctuality and for schedules and deadlines should be related to polychronicity, we included Schriber and Gutek's (1987) punctuality and schedules-and-deadlines scales in the questionnaire administered to samples 10 and 11 in order to test the IPV's discriminant validity with confirmatory factor analysis. Discriminant validity refers to the ability to differentiate a construct from other constructs that may be similar (Kerlinger, 1973, p. 462), and the kindred temporal concepts Hall specified as being related to polychronicity seemed reasonable candidates for use in a discriminant validity assessment of the IPV.

The CFAs we conducted compared a three-factor model (separate polychronicity, schedules-and-deadlines, and punctuality factors) with a two-factor model (a separate polychronicity factor and a factor that combined the schedules-and-deadlines and punctuality items on a second factor suggested by the content adequacy assessment (see Table II)) and a one-factor model (all

polychronicity, schedules-and-deadlines, and punctuality items specified as loading on a single factor). The three-factor model specified all ten IPV scale items to load on a single factor, and all of the items from Schriber and Gutek's (1987) punctuality scale and schedules-and-deadlines scale were specified to load on a second and third factor, respectively. All potential cross-loadings were prohibited in the model specifications although the three latent variables were allowed to correlate.

The results presented in Table VI indicate convergent and discriminant validity for the scale. The better fit of the three-factor models over the corresponding two- and one-factor models in both samples supports the conclusion that polychronicity is a different construct than the potentially related punctuality and schedules-and-deadlines factors, thus indicating discriminant validity. The scale's ten items also all loaded significantly on a single factor in both samples when they were analyzed by themselves and in the three-factor solutions just described, which indicates convergent validity.

The three factors were significantly intercorrelated in the three-factor model. The polychronicity factor was significantly correlated (one-tailed tests) with the schedules-and-deadlines factor in both sample 10 ($r = -0.23, p < .001$) and sample 11 ($r = -0.13, p < 0.03$), as was the punctuality factor with the schedules-and-deadlines factor ($r = 0.52, p < 0.001$; $r = 0.53, p < 0.001$, respectively); however, the polychronicity factor was not correlated significantly with the punctuality factor in either sample. When the factor correlations were set to zero, the three-factor solution still revealed a significantly better fit than the two-factor model in both samples ($\chi^2(1) = 100.51, p < 0.001$; $\chi^2(1) = 63.42, p < 0.001$, in samples 10 and 11, respectively)

Model	χ^2	df	$\Delta\chi^2$	p
<i>Sample 10</i>				
Null	5,245.22	253	NA	NA
One-factor model	3,197.41	230	2,047.81	< 0.001
Two-factor model	1,873.45	229	1,323.96	< 0.001
Three-factor model	1,583.32	228	290.13	< 0.001
<i>Sample 11</i>				
Null	1,863.87	253	NA	NA
One-factor model	923.52	230	940.35	< 0.001
Two-factor model	934.75	229	11.24	< 0.001
Three-factor model	801.11	228	133.65	< 0.001

Notes: Because of additional missing values in the punctuality and schedules-and-deadlines scales, the sample sizes changed. Sample 10 = 853 and sample 11 = 321. The object of these CFAs was to examine the IPV's ability to differentiate itself from the other scales rather than to maximize model fit, so fit indices were not emphasized. What follow are the fit indices for the CFAs in the two samples ranging in order from the one-factor model to the three-factor model: sample 10 (GFI: 0.67, 0.81, 0.83; CFI: 0.46, 0.70, 0.75; IFI: 0.46, 0.70, 0.75; SRMR: 0.13, 0.08, 0.08); sample 11 (GFI: 0.67, 0.78, 0.80; CFI: 0.46, 0.67, 0.73; IFI: 0.46, 0.67, 0.73; SRMR: 0.13, 0.09, 0.08)

Table VI.
Chi-square test of fit
for the polychronicity
(IPV), punctuality, and
schedules-and-deadlines
scales in samples 10
and 11

although the three-factor model that allowed correlations among the three latent variables revealed significantly better fit than the three-factor model with orthogonal latent variables ($\chi^2(3) = 189.62, p < 0.001$; $\chi^2(3) = 70.23, p < 0.001$, in samples 10 and 11, respectively). These analyses indicate the IPV's discriminant validity because the variance the IPV shares with its measures, the individual scale items, is greater than the variance it shares with the other constructs (punctuality and schedules-and-deadlines emphasis) with which, according to theory, it should have been related (Usunier and Valette-Florence, 1994, p. 233).

Nomological validity

Cronbach and Meehl (1955) proposed testing a nomological net to support claims for a scale's construct validity, a nomological net being the interlocking system of laws constituting a theory. They also noted that in a construct's early history, the network will be limited and the construct will have few connections. Although polychronicity can properly be described as a construct early in its history, Hall's work began a nomological net that provides predictions about the relationships among polychronicity, punctuality, and schedules-and-deadlines values at the cultural level of analysis.

The observations by Hall (1983) and Hall and Hall (1990) indicate polychronicity values should be negatively correlated with both punctuality and schedules-and-deadlines values. Specifically, the more polychronic the culture, the less punctuality will be valued and the less importance will be attached to meeting schedules and deadlines. Completing this small nomological net, punctuality values and schedules-and-deadlines values should be positively correlated.

We tested this nomological net by calculating the correlations among the temporal values of the departments in the two hospitals (samples 10 and 11). Respondents in these hospitals had also completed the IPV and Schriber and Gutek's (1987) punctuality and schedules-and-deadlines scales with their departments as the referent. Hence, in the analysis of this nomological net the respondent's department was the referent rather than the respondent's entire organization, and each department's score for each temporal value was the mean of the respondents from that department. The correlations among the three variables are presented in Table VII, all of which are significant, and more importantly, in the expected directions: punctuality and schedules-and-deadlines values were positively correlated, and both punctuality values and schedules-and-deadlines values were negatively correlated with polychronicity.

When examining group cultures – departmental cultures in this case – an important issue involves how many respondents or judges are required to make observations about each group (department). Because some of the data upon which the correlations in Table VII are based came from single observers (sometimes only one individual was included from very small departments), the correlation matrix in Table VII was recalculated nine times with increasing requirements for the number of respondents per unit. As this requirement increased from at least one respondent per unit to at least ten respondents per

department, the number of departments in the analysis declined steadily (e.g. with a requirement of at least ten respondents per department, 66 departments remained for analysis). However, the identical pattern of correlations was found in all of these analyses: punctuality and schedules-and-deadlines values were positively correlated, and both punctuality values and schedules-and-deadlines values were negatively correlated with polychronicity. Although some of the correlations in these analyses failed to attain significance at the 0.05 level, all of the correlations were significant at the 0.10 level (one-tailed tests). Thus, regardless of the minimum number of respondents per department, the expected relationships were found between both punctuality and schedules-and-deadlines values and polychronicity. These results strongly and consistently support the IPV's nomological validity.

Discussion and conclusion

Over a development process spanning 11 samples that totaled 2,190 respondents, the Inventory of Polychronic Values was supported as a valid and reliable measure of polychronicity. In addition to meeting the goal of consistently producing alpha reliability coefficients in the 0.80 range (the median of the 13 alphas is 0.84), the IPV demonstrated:

- (1) strong test-retest reliability;
- (2) little, if any, susceptibility to social desirability response biases;
- (3) content adequacy in an extremely supportive set of results based on Schriesheim *et al.*'s (1993) Q-Method and an equally supportive known groups test;
- (4) stable theoretical homogeneity in a series of confirmatory factor analyses conducted in and across three large samples;
- (5) discriminant validity by achieving a better model fit in two samples when scales measuring punctuality and schedules-and-deadlines values were included in CFAs comparing one-, two-, and three-factor models of the factor structure; and
- (6) convergent validity by all ten IPV items loading significantly on a single polychronicity factor in all of the confirmatory factor analyses conducted.

Cultural value	1	2	3
1. Polychronicity	(0.76)		
2. Punctuality	-0.22***	(0.58)	
3. Schedules-and-deadlines emphasis	-0.22***	0.39***	(0.80)

Notes: Correlations with polychronicity values are based on an N of 198 departments. The correlations between the punctuality and schedules-and-deadlines values are based on an N of 199 departments. The numbers on the diagonal are Cronbach's (1951) alpha coefficients; *** $p < 0.001$, one-tailed test

Table VII.
Correlations among
department-level
polychronicity,
punctuality and
schedules-and-deadlines
values in samples 10
and 11

Psychometrically, the IPV appears to possess the attributes of a sound (valid and reliable) scale for measuring the polychronicity construct as a dimension of organizational culture[3].

Having developed a valid and reliable measure of polychronicity, we can return to the polychronicity-related issues described at the beginning of this article to illustrate the kinds of research questions researchers can now address empirically. Is Schein's (1992) theorizing accurate about small, young organizations being polychronic whereas large, mature organizations should be more monochronic? Is polychronicity negatively correlated with organizational size and age, and if so, is organizational performance and effectiveness contingent on the proper relationships among polychronicity, size, and age? Following Eisenhardt (1989) and Judge and Miller (1991), do organizations with polychronic cultures tend to have strategic management teams that consider more alternatives during decision making than organizations with less polychronic cultures? Does number of decisions considered mediate a possible polychronicity-organizational effectiveness relationship? Given Judge and Miller's specification that many of Eisenhardt's findings appear to be limited to organizations in high velocity environments, is a polychronic culture an effective organizational response to high velocity environments? Following Brown and Eisenhardt (1997), would a highly polychronic culture promote the effective management of the simultaneous multiple-product-development process? Does McCollum and Sherman's (1991) finding that matrix organizations performed better as the percentage of personnel increased who were assigned to two projects rather than to either one or to three or more projects mean that matrix organizations with moderately polychronic cultures are more effective than matrix organizations with either extremely monochronic or extremely polychronic cultures?

Although these questions are by no means exhaustive of the fundamental organizational research issues involving polychronicity, they illustrate important issues involving polychronicity and fundamental organizational research concerns. With the development of the Inventory of Polychronic Values, these issues can now be studied directly. Some of them might even be examined simultaneously.

Notes

1. Hall's discussions reveal ambiguity about the precise definition of polychronicity (e.g. whether it includes level of emphasis on punctuality and schedules, Hall, 1976, p. 17; Hall, 1983, p. 53; Hall and Hall, 1990, pp. 13-15). He has proffered definitions or explanations that focus on preferences for involvement in multiple activities simultaneously, and we have followed other polychronicity researchers (e.g. Bluedorn *et al.*, 1992; Kaufman *et al.*, 1989, 1991; Schein, 1992; Slocombe and Bluedorn, 1999) who have defined polychronicity following the more tightly focused, preferences-for-doing-multiple-things-at-a-time discussion in Hall's work (Hall, 1959, pp. 153, 155; Hall, 1983, pp. 45-46; Hall and Hall, 1990, p. 13). Treating polychronicity as a tightly focused construct allows us to avoid the problems of information loss and ambiguity in the interpretation of results that Carver (1989) has noted as inherent in multidimensional composite measures.

2. Even though content adequacy would normally be assessed early in the scale development process, both tests of content adequacy were conducted after most of the other data had been collected and analyzed. The reason for this divergence from traditional practice is that:
 - the authors did not become aware of the Schriesheim *et al.* (1993) technique until after they had performed the other analyses; and
 - similarly, the idea of writing scenarios to perform the known-groups test did not occur to them until very late in the overall project. The content adequacy results are presented where they would normally have been located had the tests upon which they are based been performed at the traditional time in this research. Both content adequacy tests strongly supported the scale, so had we conducted them at the usual point in the research, we would have collected the other data and analyzed them as we did anyway.
3. A sizeable amount of unpublished data collected by the first author indicates that the IPV can be easily modified to provide an equally valid and reliable measure of individual-level polychronicity. To do so, the word “we” in each of the ten statements (see Tables I and III) is simply changed to “I,” and “ourselves” is changed to “myself” in item 4, which alters the referent from a group (i.e. department, organization, etc.) to the individual respondent. The instructions in the questionnaire would also be changed to direct respondents to answer the ten questions about themselves. Alpha coefficients calculated for the individual version of the IPV in these unpublished data were comparable to the alpha coefficients for the cultural-level coefficients reported in this article.

References

- Alwin, D.F. and Krosnick, J.A. (1985), “The measurement of values in surveys: a comparison of ratings and rankings”, *Public Opinion Quarterly*, Vol. 49 No. 4, pp. 535-52.
- Bentler, P.M. (1992), “On the fit of models to covariances and methodology to the bulletin”, *Psychological Bulletin*, Vol. 112 No. 3, pp. 400-4.
- Bentler, P.M. (1995), *EQS: Structural Equations Program Manual*, Multivariate Software, Inc., Los Angeles, CA.
- Bentler, P.M. and Bonett, D.G. (1980), “Significance tests and goodness-of-fit in the analysis of covariance structures”, *Psychological Bulletin*, Vol. 88 No. 3, pp. 588-606.
- Bluedorn, A.C. and Denhardt, R.B. (1988), “Time and organizations”, *Journal of Management*, Vol. 14 No. 2, pp. 299-319.
- Bluedorn, A.C., Kaufman, C.F. and Lane, P.M. (1992), “How many things do you like to do at once? An introduction to monochronic and polychronic time”, *Academy of Management Executive*, Vol. 6 No. 4, pp. 17-26.
- Brown, S.L. and Eisenhardt, K.M. (1997), “The art of continuous change: linking complexity theory and time-paced evolution in relentlessly shifting organizations”, *Administrative Science Quarterly*, Vol. 42 No. 1, pp. 1-34.
- Carver, C.S. (1989). “How should multifaceted personality constructs be tested? Issues illustrated by self-monitoring, attributional style, and hardiness”, *Journal of Personality and Social Psychology*, Vol. 56 No. 4, pp. 577-85.
- Clark, L.A. and Watson, D. (1995), “Constructing validity: basic issues in objective scale development”, *Psychological Assessment*, Vol. 7 No. 3, pp. 309-19.
- Cronbach, L.J. (1951), “Coefficient alpha and the internal structure of tests”, *Psychometrika*, Vol. 16 No. 3, pp. 297-334.
- Cronbach, L.J. and Meehl, P.E. (1955), “Construct validity in psychological tests”, *Psychological Bulletin*, Vol. 52 No. 4, pp. 281-302.
- Crowne, D.P. and Marlowe, D. (1964), *The Approval Motive*, J. Wiley & Sons, New York, NY.
- Das, T.K. (1986), *The Subjective Side of Strategy Making: Future Orientations and Perceptions of Executives*, Praeger, New York, NY.

- Eisenhardt, K.M. (1989), "Making fast strategic decisions in high-velocity environments", *Academy of Management Journal*, Vol. 32 No. 3, pp. 543-76.
- Feldt, L.S., Woodruff, D.J. and Salih, F.A. (1987), "Statistical inference for coefficient alpha", *Applied Psychological Measurement*, Vol. 11 No. 1, pp. 93-103.
- Fisher, R.A. (1921), "On the 'probable error' of a coefficient of correlation deduced from a small sample", *Metron*, Vol. 1 No. 4, pp. 3-32.
- Gorsuch, R.L. (1983), *Factor Analysis* (2nd ed.), Lawrence Erlbaum Associates, Hillsdale, NJ.
- Hall, E.T. (1959), *The Silent Language*, Anchor Books, New York, NY.
- Hall, E.T. (1976), *Beyond Culture*, Anchor Books, New York, NY.
- Hall, E.T. (1983), *The Dance of Life*, Anchor Press, Garden City, NY.
- Hall, E.T. and Hall, M.R. (1990), *Understanding Cultural Differences*, Intercultural Press, Yarmouth, ME.
- Harvey, R.J., Billings, R.S. and Nilan, K.J. (1985), "Confirmatory factor analysis of the job diagnostic survey: good news and bad news", *Journal of Applied Psychology*, Vol. 70 No. 3, pp. 461-8.
- Hoyle, R.H. and Panter, A.T. (1995), "Writing about structural equation models", in Hoyle, R.H. (Ed.), *Structural Equation Modeling: Concepts, Issues, and Applications*, Sage, Thousand Oaks, CA, pp. 158-76.
- Hu, L. and Bentler, P.M. (1995), "Evaluating model fit", in Hoyle, R.H. (Ed.), *Structural Equation Modeling: Concepts, Issues, and Applications*, Sage, Thousand Oaks, CA, pp. 76-99.
- Judge, W.Q. and Miller, A. (1991), "Antecedents and outcomes of decision speed in different environmental contexts", *Academy of Management Journal*, Vol. 34 No. 2, pp. 449-63.
- Kaufman, C.F., Lane, P.M. and Lindquist, J.D. (1989), "Exploring more than 24 hours a day: a preliminary investigation of polychronic time use", Paper presented at the Association for Consumer Research, New Orleans, LA, December.
- Kaufman, C.F., Lane, P.M. and Lindquist, J.D. (1991), "Exploring more than 24 hours a day: a preliminary investigation of polychronic time use", *Journal of Consumer Research*, Vol. 18 No. 3, pp. 392-401.
- Kerlinger, F.N. (1973), *Foundations of Behavioral Research* (2nd ed.), Holt, Rinehart & Winston, New York, NY.
- Kotter, J.P. and Heskett, J.L. (1992), *Corporate Culture and Performance*, Free Press, New York, NY.
- Kroeber, A.L. and Kluckhohn, C. (1952), *Culture: A Critical Review of Concepts and Definitions, Papers of the Peabody Museum of American Archaeology and Ethnology*, Vol. 47 No. 1, Harvard University, Cambridge, MA.
- Lautenschlager, G.J. (1989), "ALPHATST: testing for differences in values of coefficient alpha", *Applied Psychological Measurement*, Vol. 13 No. 3, p. 284.
- McCollum, J.K. and Sherman, J.D. (1991), "The effects of matrix organization size and number of project assignments on performance", *IEEE Transactions on Engineering Management*, Vol. 38 No. 1, pp. 75-77.
- McGrath, J.E. and Rotchford, N.L. (1983), "Time and behavior in organizations", *Research in Organizational Behavior*, Vol. 5, pp. 57-101.
- Nunnally, J.C. (1978), *Psychometric Theory*, McGraw-Hill, New York, NY.
- O'Reilly, C.A. III, Chatman, J. and Caldwell, D.F. (1991), "People and organizational culture: a profile comparison approach to assessing person-organization fit", *Academy of Management Journal*, Vol. 34 No. 3, pp. 487-516.
- Ouchi, W.G. (1981), *Theory Z*, Avon Books, New York, NY.
- Peters, T.J. and Waterman, R.H. Jr (1982), *In Search of Excellence*, Harper & Row, New York, NY.

- Quinn, R.E. and Cameron, K. (1983), "Organizational life cycles and shifting criteria of organizational effectiveness: some preliminary evidence", *Management Science*, Vol. 29 No. 1, pp. 33-51.
- Robinson, J.B., Shaver, P.R. and Wrightsman, L.S. (1991), *Measures of Personality and Social Psychological Attitudes*, Academic Press, Inc., San Diego, CA.
- Rosenthal, R. and Rosnow, R.L. (1991), *Essentials of Behavioral Research* (2nd ed.), McGraw-Hill, New York, NY.
- Schein, E.H. (1985), *Organizational Culture and Leadership*, Jossey-Bass, San Francisco, CA.
- Schein, E.H. (1992), *Organizational Culture and Leadership* (2nd ed.), Jossey-Bass, San Francisco, CA.
- Schriber, J.B. and Gutek, B.A. (1987), "Some time dimensions of work: measurement of an underlying aspect of organizational culture", *Journal of Applied Psychology*, Vol. 72 No. 4, pp. 642-50.
- Schriesheim, C.A., Powers, K.J., Scandura, T.A., Gardiner, C.C. and Lankau, M.J. (1993), "Improving construct measurement in management research: comments and a quantitative approach for assessing the theoretical content adequacy of paper-and-pencil survey-type instruments", *Journal of Management*, Vol. 19 No. 2, pp. 385-417.
- Sheridan, J.E. (1992), "Organizational culture and employee retention", *Academy of Management Journal*, Vol. 35 No. 5, pp. 1036-56.
- Slocombe, T.E. and Bluedorn, A.C. (1999), "Organizational behavior implications of the congruence between preferred polychronicity and experienced work-unit polychronicity", *Journal of Organizational Behavior*, Vol. 20 No. 1, pp. 75-99.
- Spector, P.E., Van Katwyk, P.T., Brannick, M.T. and Chen, P.Y. (1997), "When two factors don't reflect two constructs: how item characteristics can produce artifactual factors", *Journal of Management*, Vol. 23 No. 5, pp. 659-77.
- Stephenson, W. (1953), *The Study of Behavior: Q-technique and Its Methodology*, University of Chicago Press, Chicago, IL.
- Usunier, J.C. and Valette-Florence, P. (1994), "Perceptual time patterns ('time styles'): a psychometric scale", *Time and Society*, Vol. 3 No. 2, pp. 219-41.

Appendix

The high polychronicity and low polychronicity scenarios

The low-level of polychronicity scenario. Please carefully read the following description of managerial work at organization A. After you finish reading the description, please answer the ten questions on the next page in the same way that you believe Jane and Tom would answer them. Jane and Tom are two managers who work at organization A, and their behaviors are included in the following description of managerial work at organization A.

Description of managerial work at organization A

Jane began her workday by inspecting the production line. As she inspected the line, one worker asked Jane to explain the company's retirement policy. Jane said: "Please call my secretary and make an appointment to see me about that later. I'm inspecting the production line now". Jane returned to her inspection and received a call on her cellular phone. The call was from one of the company's sales representatives, and the representative wanted some information about a product manufactured in Jane's unit. Jane asked the representative to call her secretary for an appointment to discuss the matter and returned to her inspection of the production line. As Jane continued her inspection, she noticed a machine had been left running, which created an extremely dangerous safety hazard. So Jane stopped her inspection, turned off the machine, and then resumed the inspection. Before she finally finished her inspection, Jane received questions from two other workers, both of whom she referred to her secretary to make appointments.

While Jane was performing the tasks described above, Tom, the head of the Human Resources Department, was giving a plant tour to several new college graduates the company

wanted to hire. He pointed Jane out to them from a distance and said, "Jane is one of our best managers. She provides an excellent example of the way we want every manager to do things here. Watch how she works, because if you join us, we will want you to handle things just like Jane does. Our CEO thinks Jane will be president of the company someday."

After returning to her office, Jane examined her to-do list for the day and found that it contained the following five items: (1) developing a new performance appraisal form for her unit; (2) writing a letter of recommendation for one of her current employees; (3) preparing an oral presentation she would be making next week; (4) writing a request for a new machine; and (5) reviewing the résumés of three job candidates.

Jane picked the task of writing a request for a new machine and worked on it for the rest of the morning, neither working on nor thinking about the other four projects. Jane received eight phone calls during the morning, all of which were taken by her secretary who made appointments with the callers for times later in the week when they should call back. Jane made significant progress on the new machine proposal and completed it just before lunch. The other four projects, however, remained untouched as the afternoon began.

The high level of polychronicity scenario

Please carefully read the following description of managerial work at Organization B. After you finish reading the description, please answer the ten questions on the next page in the same way that you believe Jane and Tom would answer them. Jane and Tom are two managers who work at Organization B, and their behaviors are included in the following description of managerial work at Organization B.

Description of Managerial Work at Organization B

Jane began her workday by inspecting the production line. As she inspected the line, one worker asked Jane to explain the company's retirement policy. Jane said, "Walk with me while I inspect the line, and I will explain the policy to you." After explaining the retirement policy, Jane received a call on her cellular phone. The call was from one of the company's sales representatives, and the representative wanted some information about a product manufactured in Jane's unit. Jane began providing the information as she continued her inspection of the production line. As Jane continued both her conversation with the sales representative and her inspection, she noticed a machine had been left running, which created an extremely dangerous safety hazard. So Jane turned off the machine while continuing her conversation with the sales representative. Before she finished her inspection, Jane received two more questions from workers, both of which she answered while continuing her inspection of the production line.

While Jane was performing the tasks described above, Tom, the head of the Human Resources Department, was giving a plant tour to several new college graduates the company wanted to hire. He pointed Jane out to them from a distance and said, "Jane is one of our best managers. She provides an excellent example of the way we want every manager to do things here. Watch how she works, because if you join us, we will want you to handle things just like Jane does. Our CEO thinks Jane will be President of the company someday."

After returning to her office, Jane examined her to-do list for the day, and she spent the rest of the morning working on the five projects it contained: (1) developing a new performance appraisal form for her unit; (2) writing a letter of recommendation for one of her employees; (3) preparing an oral presentation she would be making next week; (4) writing a request for a new machine; and (5) reviewing the résumés of three job candidates. Jane often developed ideas for the other projects while she worked on one of them, so she frequently switched back-and-forth among the projects throughout the morning. Jane received eight phone calls during the morning, and she continued working on the different projects as she spoke on the phone. Jane made significant progress on several of the projects during the morning, but more work was needed on all of the projects as the afternoon began.

This article has been cited by:

1. Ewa Szumowska, Małgorzata Kossowska. 2017. Motivational rigidity enhances multitasking performance: The role of handling interruptions. *Personality and Individual Differences* **106**, 81-89. [[CrossRef](#)]
2. Lin Lin, Leila A. Mills, Dirk Ifenthaler. 2016. Collaboration, multi-tasking and problem solving performance in shared virtual spaces. *Journal of Computing in Higher Education* **28**:3, 344-357. [[CrossRef](#)]
3. Jinwoo Kim, Jinkyu Jang, Inseong Lee. 2016. The Effects of Hover Interface on Users' Behavioral Multitasking Intention. *International Journal of Human-Computer Interaction* . [[CrossRef](#)]
4. Catarina Marques Santos, Ana Margarida Passos, Sjir Uitdewilligen, Annika Nübold. 2016. Shared temporal cognitions as substitute for temporal leadership: An analysis of their effects on temporal conflict and team performance. *The Leadership Quarterly* **27**:4, 574-587. [[CrossRef](#)]
5. Mustafa Daskin School of Tourism and Hotel Management, Sinop University, Sinop, Turkey . 2016. Linking polychronicity to hotel frontline employees' job outcomes. *EuroMed Journal of Business* **11**:2, 162-180. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
6. Zhongju Liao. 2016. Temporal cognition, environmental innovation, and the competitive advantage of enterprises. *Journal of Cleaner Production* . [[CrossRef](#)]
7. Mustafa Daskin School of Tourism and Hospitality Management, Sinop University, Sinop, Turkey Ozlem Altunoz Surucu School of Tourism and Hospitality Management, Sinop University, Sinop, Turkey . 2016. The role of polychronicity and intrinsic motivation as personality traits on frontline employees' job outcomes. *Asia-Pacific Journal of Business Administration* **8**:2, 177-196. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
8. Yubo Zhang, Pei-Luen Patrick Rau. 2016. An Exploratory Study to Measure Excessive Involvement in Multitasking Interaction with Smart Devices. *Cyberpsychology, Behavior, and Social Networking* **19**:6, 397-403. [[CrossRef](#)]
9. Karen Korabik, Tricia van Rhijn, Roya Ayman, Donna S. Lero, Leslie B. Hammer. 2016. Gender, polychronicity, and the work-family interface: is a preference for multitasking beneficial?. *Community, Work & Family* 1-20. [[CrossRef](#)]
10. Mihye Seo, Jung-Hyun Kim, Prabu David. 2015. Always Connected or Always Distracted? ADHD Symptoms and Social Assurance Explain Problematic Use of Mobile Phone and Multicommunicating. *Journal of Computer-Mediated Communication* **20**:6, 667-681. [[CrossRef](#)]
11. Mustafa Daskin. 2015. Antecedents of extra-role customer service behaviour: polychronicity as a moderator. *Anatolia* **26**:4, 521-534. [[CrossRef](#)]
12. Jie Zhang Department of Communication, University of Evansville, Evansville, Indiana, USA Wei-Na Lee Department of Advertising, University of Texas at Austin, Austin, Texas, USA . 2015. Testing the concepts of market mavenism and opinion leadership in China. *American Journal of Business* **30**:3, 178-195. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
13. Josette Gevers, Susan Mohammed, Nataliya Baytalskaya. 2015. The Conceptualisation and Measurement of Pacing Styles. *Applied Psychology* **64**:3, 499-540. [[CrossRef](#)]
14. Elisa Mattarelli, Fabiola Bertolotti, Valerio Incerti. 2015. The interplay between organizational polychronicity, multitasking behaviors and organizational identification: A mixed-methods study in knowledge intensive organizations. *International Journal of Human-Computer Studies* **79**, 6-19. [[CrossRef](#)]
15. Georgia Kouali Department of Education, Open University of Cyprus, Nicosia, Cyprus Petros Pashiardis Department of Education, Open University of Cyprus, Nicosia, Cyprus . 2015. Time management profiles

of Cypriot school principals: a mixed-methods approach. *International Journal of Educational Management* 29:4, 492-518. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]

16. Fabiola Bertolotti, Elisa Mattarelli, Matteo Vignoli, Diego Maria Macrì. 2015. Exploring the relationship between multiple team membership and team performance: The role of social networks and collaborative technology. *Research Policy* 44:4, 911-924. [[CrossRef](#)]
17. Abbie J. Shipp, Michael S. Cole. 2015. Time in Individual-Level Organizational Studies: What Is It, How Is It Used, and Why Isn't It Exploited More Often?. *Annual Review of Organizational Psychology and Organizational Behavior* 2:1, 237-260. [[CrossRef](#)]
18. Gloria Mark. 2015. Multitasking in the Digital Age. *Synthesis Lectures on Human-Centered Informatics* 8:3, 1-113. [[CrossRef](#)]
19. Neus Capdeferro, Margarida Romero, Elena Barberà. 2014. Polychronicity: review of the literature and a new configuration for the study of this hidden dimension of online learning. *Distance Education* 35:3, 294-310. [[CrossRef](#)]
20. Huseyin Arasli, Mustafa Daşkın, Serdar Saydam. 2014. Polychronicity and Intrinsic Motivation as Dispositional Determinants on Hotel Frontline Employees' Job Satisfaction: Do Control Variables Make a Difference?. *Procedia - Social and Behavioral Sciences* 109, 1395-1405. [[CrossRef](#)]
21. Susan Mohammed, David A. Harrison. 2013. The clocks that time us are not the same: A theory of temporal diversity, task characteristics, and performance in teams. *Organizational Behavior and Human Decision Processes* 122:2, 244-256. [[CrossRef](#)]
22. Helena M. AddaeCollege of Business and Economics, University of Wisconsin-Whitewater, Whitewater, Wisconsin, USA Gary JohnsJohn Molson School of Business, Concordia University, Montreal, Canada Kathleen BoiesJohn Molson School of Business, Concordia University, Montreal, Canada. 2013. The legitimacy of absenteeism from work: a nine nation exploratory study. *Cross Cultural Management: An International Journal* 20:3, 402-428. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
23. Karim Babayi Nadinloyi, Nader Hajloo, Nasser Sobhi Garamaleki, Hasan Sadeghi. 2013. The Study Efficacy of Time Management Training on Increase Academic Time Management of Students. *Procedia - Social and Behavioral Sciences* 84, 134-138. [[CrossRef](#)]
24. Ann-Frances Cameron, Jane Webster. 2013. Multicommunicating: Juggling Multiple Conversations in the Workplace. *Information Systems Research* 24:2, 352-371. [[CrossRef](#)]
25. Antonio Pierro, Mauro Giacomantonio, Gennaro Pica, Arie W. Kruglanski, E. Tory Higgins. 2013. Locomotion and the preference for multi-tasking: Implications for well-being. *Motivation and Emotion* 37:2, 213-223. [[CrossRef](#)]
26. Christophe Fournier, William A. Weeks, Christopher P. Blocker, Lawrence B. Chonko. 2013. Polychronicity and Scheduling's Role in Reducing Role Stress and Enhancing Sales Performance. *Journal of Personal Selling and Sales Management* 33:2, 197-210. [[CrossRef](#)]
27. Tejinder K. BillingDepartment of Management and Entrepreneurship, Rohrer College of Business, Rowan University, Glassboro, New Jersey, USA Rabi S. BhagatDepartment of Management, University of Memphis, Memphis, Tennessee, USA Emin BabakusDepartment of Marketing and Supply Chain Management, University of Memphis, Memphis, Tennessee, USA. 2013. Task structure and work outcomes. *Management Research Review* 36:2, 136-152. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
28. Karen J. Jansen, Abbie J. ShippA Review and Agenda for Incorporating Time in Fit Research 195-221. [[CrossRef](#)]

29. Osman M. Karatepe, Georgiana Karadas, Arezou Khabbaz Azar, Nazanin Naderiadib. 2013. Does Work Engagement Mediate the Effect of Polychronicity on Performance Outcomes? A Study in the Hospitality Industry in Northern Cyprus. *Journal of Human Resources in Hospitality & Tourism* 12:1, 52-70. [[CrossRef](#)]
30. Matthew J. Grawitch, Larissa K. Barber. 2013. In Search of the Relationship Between Polychronicity and Multitasking Performance. *Journal of Individual Differences* 34:4, 222-229. [[CrossRef](#)]
31. Cross-Cultural Psychology 5-24. [[CrossRef](#)]
32. Jichul Jang, R. Thomas George. 2012. Understanding the influence of polychronicity on job satisfaction and turnover intention: A study of non-supervisory hotel employees. *International Journal of Hospitality Management* 31:2, 588-595. [[CrossRef](#)]
33. Yan Luximon, Ravindra S. Goonetilleke. 2012. Time use behavior in single and time-sharing tasks. *International Journal of Human-Computer Studies* 70:5, 332-345. [[CrossRef](#)]
34. Tracy M. Kantrowitz, Darrin M. Grelle, James C. Beaty, Mark B. Wolf. 2012. Time Is Money: Polychronicity as a Predictor of Performance Across Job Levels. *Human Performance* 25:2, 114-137. [[CrossRef](#)]
35. Brett Agypt, Beth A. Rubin. 2012. Time in the New Economy: The Impact of the Interaction of Individual and Structural Temporalities on Job Satisfaction. *Journal of Management Studies* 49:2, 403-428. [[CrossRef](#)]
36. Etty G.A. Wielenga-Meijer, Toon W. Tàris, Daniël H.J. Wigboldus, Michiel A.J. Kompier. 2012. Don't bother me: learning as a function of task autonomy and cognitive demands. *Human Resource Development International* 15:1, 5-23. [[CrossRef](#)]
37. Keri K. Stephens, Jaehae K. Cho, Dawna I. Ballard. 2012. Simultaneity, Sequentiality, and Speed: Organizational Messages About Multiple-Task Completion. *Human Communication Research* 38:1, 23-47. [[CrossRef](#)]
38. Katherine Richardson, Raquel Benbunan-Fich. 2011. Examining the antecedents of work connectivity behavior during non-work time. *Information and Organization* 21:3, 142-160. [[CrossRef](#)]
39. Premjit K. Sanjram, Azizuddin Khan. 2011. Attention, polychronicity, and expertise in prospective memory performance: Programmers' vulnerability to habit intrusion error in multitasking. *International Journal of Human-Computer Studies* 69:6, 428-439. [[CrossRef](#)]
40. Ann-Frances Cameron, Jane Webster. 2011. Relational Outcomes of Multicommunicating: Integrating Incivility and Social Exchange Perspectives. *Organization Science* 22:3, 754-771. [[CrossRef](#)]
41. Han Li, Ashish Gupta, Xin Luo, Merrill Warkentin. 2011. Exploring the impact of instant messaging on subjective task complexity and user satisfaction. *European Journal of Information Systems* 20:2, 139-155. [[CrossRef](#)]
42. C Ritterskamp The Collaborative Nature of Interruption Handling 1-10. [[CrossRef](#)]
43. Simon J.M. Adams Facultad de Derecho, Universidad de San Pablo CEU, Madrid, Spain CUNEF, Universidad Complutense de Madrid, Madrid, Spain Wendelien van Eerde Amsterdam Business School, Amsterdam, The Netherlands. 2010. Time use in Spain: is polychronicity a cultural phenomenon?. *Journal of Managerial Psychology* 25:7, 764-776. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
44. Eric Chong, Xiaofang Ma. 2010. The Influence of Individual Factors, Supervision and Work Environment on Creative Self-Efficacy. *Creativity and Innovation Management* 19:3, 233-247. [[CrossRef](#)]
45. Elizabeth M. Poposki, Frederick L. Oswald. 2010. The Multitasking Preference Inventory: Toward an Improved Measure of Individual Differences in Polychronicity. *Human Performance* 23:3, 247-264. [[CrossRef](#)]

46. Cornelius J. König, Mary J. Waller. 2010. Time for Reflection: A Critical Examination of Polychronicity. *Human Performance* 23:2, 173-190. [[CrossRef](#)]
47. Cornelius J. König, Lourdes Oberacher, Martin Kleinmann. 2010. Personal and Situational Determinants of Multitasking at Work. *Journal of Personnel Psychology* 9:2, 99-103. [[CrossRef](#)]
48. William A. Weeks, Christophe Fournier. 2009. The Impact of Time Congruity on Salesperson's Role Stress: A Person-Job Fit Approach. *Journal of Personal Selling and Sales Management* 30:1, 73-90. [[CrossRef](#)]
49. Joshua M. DavisCollege of Charleston Lorraine S. LeeUniversity of North Carolina -Wilmington Mun Y. YiUniversity of South Carolina. 2009. Time-User Preference and Technology Acceptance: Measure Development of Computer Polychronicity. *American Journal of Business* 24:2, 23-32. [[Abstract](#)] [[PDF](#)]
50. Amy Todd. 2009. From Polychronicity to Multitasking: The Warping of Time Across Disciplinary Boundaries. *Anthropology of Work Review* 30:2, 49-54. [[CrossRef](#)]
51. Abbie J. Shipp, Jeffrey R. Edwards, Lisa Schurer Lambert. 2009. Conceptualization and measurement of temporal focus: The subjective experience of the past, present, and future. *Organizational Behavior and Human Decision Processes* 110:1, 1-22. [[CrossRef](#)]
52. Rachel L. Neo, Marko M. Skoric. 2009. Problematic Instant Messaging Use. *Journal of Computer-Mediated Communication* 14:3, 627-657. [[CrossRef](#)]
53. Vangelis Souitaris, B. M. Marcello Maestro. 2009. Polychronicity in top management teams: The impact on strategic decision processes and performance of new technology ventures. *Strategic Management Journal* n/a-n/a. [[CrossRef](#)]
54. G.T. Lumpkin, Claudia C. Coglisser, Dawn R. Schneider. 2009. Understanding and Measuring Autonomy: An Entrepreneurial Orientation Perspective. *Entrepreneurship Theory and Practice* 33:1, 47-69. [[CrossRef](#)]
55. Wendi L. Adair, Nancy R. Buchan, Xiao-Ping ChenConceptualizing Culture as Communication in Management and Marketing Research 146-180. [[CrossRef](#)]
56. Susan Mohammed, Tracey Rizzuto, Nathan J. Hiller, Daniel A. Newman, Tina Chen. 2008. Individual Differences and Group Negotiation: The Role of Polychronicity, Dominance, and Decision Rule. *Negotiation and Conflict Management Research* 1:3, 282-307. [[CrossRef](#)]
57. David A. GriffithMichael G. HarveyUniversity of Mississippi, Holman, Mississippi, USA Bond University, Gold Coast, Australia Timothy S. KiesslingBond University, Gold Coast, Australia R. Glenn RicheyYoung J. Boozer Faculty Fellow, Culverhouse College of Commerce and Business Administration, The University of Alabama, Tuscaloosa, Alabama, USA. 2008. Global social time perspectives in marketing: a strategic reference point theory application. *International Marketing Review* 25:2, 146-165. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
58. Allen C. Bluedorn, Gwen Martin. 2008. The time frames of entrepreneurs. *Journal of Business Venturing* 23:1, 1-20. [[CrossRef](#)]
59. Kraig L. Schell, Jeffrey M. Conte. 2008. Associations among polychronicity, goal orientation, and error orientation. *Personality and Individual Differences* 44:1, 288-298. [[CrossRef](#)]
60. Beth A. Rubin Time-Work Discipline in the 21st Century 1-26. [[Abstract](#)] [[Full Text](#)] [[PDF](#)] [[PDF](#)]
61. Beth A. Rubin New Times Redux: Layering Time in the New Economy 527-548. [[Abstract](#)] [[Full Text](#)] [[PDF](#)] [[PDF](#)]
62. Miri Lerner, Shaker A. Zahra, Yael Gal Kohavi Time and Corporate Entrepreneurship 187-221. [[Abstract](#)] [[Full Text](#)] [[PDF](#)] [[PDF](#)]

63. Allen C. Bluedorn Polychronicity, Individuals, and Organizations 179-222. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
64. Allen C. Bluedorn, Kimberly S. Jaussi Organizationally Relevant Dimensions of Time Across Levels of Analysis 187-223. [[Abstract](#)] [[Full Text](#)] [[PDF](#)] [[PDF](#)]
65. Jeffrey M. Conte Measuring Temporal Constructs Across Multiple Levels of Analysis 225-237. [[Abstract](#)] [[Full Text](#)] [[PDF](#)] [[PDF](#)]
66. Allen C. Bluedorn, Kimberly S. Jaussi Time and the Challenge of Temporal Concepts 249-255. [[Abstract](#)] [[Full Text](#)] [[PDF](#)] [[PDF](#)]
67. Cynthia Lee, Chun Hui, Catherine H Tinsley, Xiongying Niu. 2006. Goal orientations and performance: role of temporal norms. *Journal of International Business Studies* 37:4, 484-498. [[CrossRef](#)]
68. Nora Madjar, Greg R. Oldham. 2006. Task Rotation and Polychronicity: Effects on Individuals' Creativity. *Human Performance* 19:2, 117-131. [[CrossRef](#)]
69. Aaron Arndt, Todd J. Arnold, Timothy D. Landry. 2006. The effects of polychronic-orientation upon retail employee satisfaction and turnover. *Journal of Retailing* 82:4, 319-330. [[CrossRef](#)]
70. Tracy D. Hecht, Natalie J. Allen. 2005. Exploring links between polychronicity and well-being from the perspective of person-job fit: Does it matter if you prefer to do only one thing at a time?. *Organizational Behavior and Human Decision Processes* 98:2, 155-178. [[CrossRef](#)]
71. Waipeng Lee, Trevor M. K. Tan, Shahiraa Shahul Hameed. 2005. Polychronicity, the Internet, and the Mass Media: A Singapore Study. *Journal of Computer-Mediated Communication* 11:1, 300-316. [[CrossRef](#)]
72. Jeffrey M. Conte, Jeremy N. Gintoft. 2005. Polychronicity, Big Five Personality Dimensions, and Sales Performance. *Human Performance* 18:4, 427-444. [[CrossRef](#)]
73. S.A. Nonis, J.K. Teng, C.W. Ford. 2005. A cross-cultural investigation of time management practices and job outcomes. *International Journal of Intercultural Relations* 29:4, 409-428. [[CrossRef](#)]
74. Karen J. Jansen, Amy L. Kristof-Brown. 2005. Marching to the beat of a different drummer: Examining the impact of pacing congruence. *Organizational Behavior and Human Decision Processes* 97:2, 93-105. [[CrossRef](#)]
75. Cornelius J. Konig, Markus Buhner, Gesine Murling. 2005. Working Memory, Fluid Intelligence, and Attention Are Predictors of Multitasking Performance, but Polychronicity and Extraversion Are Not. *Human Performance* 18:3, 243-266. [[CrossRef](#)]
76. Yan Zhang, Ravindra S. Goonetilleke, Thomas Plocher, Sheau-Farn Max Liang. 2005. Time-related behaviour in multitasking situations. *International Journal of Human-Computer Studies* 62:4, 425-455. [[CrossRef](#)]
77. Vladimir Kvassov. 2004. The Effects of Time Personality on the Productivity of MIS. *Journal of Decision Systems* 13:1, 9-25. [[CrossRef](#)]
78. June Cotte Assistant Professor, The Richard Ivey School of Business, University of Western Ontario, London, Ontario, Canada S. Ratneshwar Professor of Marketing, University of Connecticut, Storrs, Connecticut, USA. 2003. Choosing leisure services: the effects of consumer timestyle. *Journal of Services Marketing* 17:6, 558-572. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
79. Alvin Hwang Department of Management and Management Science, Lubin School of Business, Pace University, New York, NY, USA. 2003. Adventure learning: competitive (Kiasu) attitudes and teamwork. *Journal of Management Development* 22:7, 562-578. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]

80. Jeffrey M. Conte, Rick R. Jacobs. 2003. Validity Evidence Linking Polychronicity and Big Five Personality Dimensions to Absence, Lateness, and Supervisory Performance Ratings. *Human Performance* 16:2, 107-129. [[CrossRef](#)]
81. V. Kvassov The effects of time and personality on the productivity of management information systems 10 pp.. [[CrossRef](#)]
82. Kyoko Ishizaka, Sandra P. Marshall, Jeffrey M. Conte. 2001. Individual Differences in Attentional Strategies in Multitasking Situations. *Human Performance* 14:4, 339-358. [[CrossRef](#)]
83. D. Lynne Persing ESC Recherche, École Supérieure de Commerce de Toulouse, Toulouse, France. 1999. Managing in polychronic times. *Journal of Managerial Psychology* 14:5, 358-373. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
84. Jeffrey M. Conte San Diego State University, San Diego, California, USA Tracey E. Rizzuto Pennsylvania State University, University Park, Pennsylvania, USA and Dirk D. Steiner Laboratoire de Psychologie Expérimentale et Quantitative, Université de Nice-Sophia Antipolis, France. 1999. A construct-oriented analysis of individual-level polychronicity. *Journal of Managerial Psychology* 14:3/4, 269-288. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
85. Marina H. Onken Florida Gulf Coast University, Fort Myers, Florida, USA. 1999. Temporal elements of organizational culture and impact on firm performance. *Journal of Managerial Psychology* 14:3/4, 231-244. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
86. Charles Benabou Université du Québec à Montréal, Canada. 1999. Polychronicity and temporal dimensions of work in learning organizations. *Journal of Managerial Psychology* 14:3/4, 257-270. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
87. David K. Palmer University of Nebraska at Kearney, Nebraska, USA F. David Schoorman Purdue University, West Lafayette, Indiana, USA. 1999. Unpackaging the multiple aspects of time in polychronicity. *Journal of Managerial Psychology* 14:3/4, 323-345. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
88. Thomas E. Slocombe University of Central Oklahoma, Edmond, Oklahoma, USA. 1999. Applying the theory of reasoned action to the analysis of an individual's polychronicity. *Journal of Managerial Psychology* 14:3/4, 313-324. [[Abstract](#)] [[Full Text](#)] [[PDF](#)]
89. Gregory M. Rose, Carina DeVilliers, Detmar Straub Chronism Theory, Culture, and System Delay 203-227. [[CrossRef](#)]
90. Lin Lin, Patricia Cranton, Jennifer Lee Research Methodologies for Multitasking Studies 329-348. [[CrossRef](#)]
91. Henrique M. G. Martins Critical Concepts in M-Health Technology Development 140-150. [[CrossRef](#)]